Medhat Elmorsy (He/Him)

Postdoctoral Fellow
Department of Civil and Environmental Engineering
Massachusetts Institute of Technology
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RESEARCH EXPERIENCE

Massachusetts Institute of Technology — Cambridge, MA
Postdoctoral Fellow — Apr. 2026 – Present

- Developing topology optimization algorithms for seismic design
- Manufacturing optimized designs using additive manufacturing and robotic techniques
- Structural testing of optimized designs

ETH Zurich — Zurich, Switzerland
Postdoctoral Researcher — Jan. 2026 – Mar. 2026

- Continued doctoral research on experimental-numerical frameworks for validation of numerical models

Scientific Assistant (Doctoral Student) — Jan. 2022 – Dec. 2025

- Proposed a novel experimental methodology combining additive manufacturing and centrifuge shake table testing to generate experimental benchmark datasets for validation of numerical models
- Integrated experimental testing, numerical (OpenSees) simulations, and probabilistic analysis to study system-level seismic behavior and uncertainty propagation

EDUCATION

ETH Zurich, Ph.D. — Zurich, Switzerland
Dissertation: System-Level Uncertainty in Earthquake Engineering: Numerical Investigation and AM-Based Physical Modeling for Benchmark Data Generation
Advisor: Prof. Dr. Michalis Vassiliou — 2025

University of Alaska Anchorage, Master's Degree — Anchorage, AK
Thesis: Nonlinear Modeling Parameters for Beam-Column Joints in Seismic Analysis of Concrete Buildings
Advisor: Prof. Dr. Wael M. Hassan — 2020

Mansoura University, Bachelor's Degree — Mansoura, Egypt
Specialization: Civil Engineering — 2016

GRANTS & AWARDS

Swiss National Science Foundation Postdoc.Mobility Fellowship — 2026

- **(~USD 227K)** postdoctoral funding for 2 years
- **Project:** Advanced Seismic Topology Optimization and Physical Prototyping of Reinforced Concrete Structures Using Additive Manufacturing and Robotic Techniques

First-Class Honours (Ranked 1st in graduating class), B.Sc. in Civil Engineering, Mansoura University — 2016

JOURNAL PUBLICATIONS (PUBLISHED)

[J12] **Elmorsy M**, Jones L, Wild M, Tatsis K, Katsamakas A, Anastasopoulos I, and Vassiliou M. Benchmark Dataset from Centrifuge Shake Table Tests of 3D-Printed URM Structures for Seismic Model Validation. *Earthquake Spectra*. 2026; 42(1): p.e70042. doi: 10.1002/esp4.70042

[J11] **Elmorsy M**, Adamopoulou D, Zhang Y, Shea K, Jones L, Anastasopoulos I, and Vassiliou M. Shear Behavior and Centrifuge Seismic Modeling of a Sustainable Corbel Dwelling with Interlocking Blocks. *Construction and Building Materials*. 2026; 515: p.145553. doi: 10.1016/j.conbuildmat.2026.145553

[J10] **Elmorsy M** and Vamvatsikos D. Impact of Beam-Column Joint Modeling Uncertainties on the Seismic Response of Low-Rise RC Frames. *Earthquake Engineering & Structural Dynamics*. 2025; 54(12): p.e70024. doi: 10.1002/eqe.70024

[J9] **Elmorsy M**, Leinenbach C, and Vassiliou MF. Cyclic Behavior of Physical Models of Reinforced Concrete Columns Built with Additive Manufacturing. *Journal of Structural Engineering*. 2025; 151(5): p.04025043. doi: 10.1061/JSENDH.STENG-13702

[J8] **Elmorsy M**, Leinenbach C, and Vassiliou MF. Small Scale Physical Modelling of Reinforced Concrete Joints Using Additively Manufactured Reinforcement. *Journal of Structural Engineering*. 2025; 151(5): p.04025036. doi: 10.1061/JSENDH.STENG-13885

[J7] **Elmorsy M**, Wrobel R, Leinenbach C, and Vassiliou MF. Additively Manufactured Steel Reinforcement for Small Scale Reinforced Concrete Modeling: Tensile and Bond Behavior. *Materials and Design*. 2024; 241: p.112919. doi: 10.1016/j.matdes.2024.112919

[J6] **Elmorsy M** and Vassiliou MF. Effect of Ground Motion Processing and Filtering on the Response of Rocking Structures. *Earthquake Engineering & Structural Dynamics*. 2023; 52(6): 1704-1721. doi: 10.1002/eqe.3837

[J5] Hassan WM and **Elmorsy M**. Cyclic Nonlinear Modeling Parameters for Unconfined Beam-Column Joints. *ACI Structural Journal*. 2022; 119(1): 89-104. doi: 10.14359/51733139

[J4] Hassan WM and **Elmorsy M**. Probabilistic Beam–Column Joint Model for Seismic Analysis of Concrete Frames. *Journal of Structural Engineering*. 2022; 148(4): p.04022011. doi: 10.1061/(ASCE)ST.1943-541X.0003235

[J3] **Elmorsy M** and Hassan WM. Seismic Behavior of Ultra-High Performance Concrete Elements: State-of-the-Art Review and Test Data Trends. *Journal of Building Engineering*. 2021; 40: p.102572. doi: 10.1016/j.jobe.2021.102572

[J2] Hassan WM and **Elmorsy M**. Database Trends and Critical Review of Seismic Performance Tests on High Strength Steel Reinforced Concrete Components. *Engineering Structures*. 2021; 239: p.112092. doi: 10.1016/j.engstruct.2021.112092

[J1] **Elmorsy M**, Ghaleb AA, and El-Zoughiby ME. Performance-Based Seismic Assessment of Buildings with Curved Shear Walls. *Engineering Research Journal*. 2019; 164: 178-192. doi: 10.21608/ERJ.2019.131347

JOURNAL PUBLICATIONS (UNDER REVIEW)

[U1] **Elmorsy M,** Jones L, Tatsis K, Katsamakas A, Anastasopoulos I, Gunay S, Mosalam KM, Kagermanov A, Wang A, Malomo D, Serrano JMT, Negrette P, Fabra J, Ghezelbash A, Aşıkoğlu A, D'Altri AM, Sharma S, Messali F, Reyes SI, and Vassiliou MF. Blind Prediction Contest on Shake Table Tests of a Masonry Model: Results with and without Structural Identification. *Earthquake Engineering & Structural Dynamics*.

JOURNAL PUBLICATIONS (IN PREPARATION)

[P3] **Elmorsy M** and Carstensen JV. A Bimodular Formulation for Topology Optimization of Reinforced Concrete Structures Under Nonlinear Seismic Demand. *Computer Methods in Applied Mechanics and Engineering.*

[P2] **Elmorsy M**, Jewett JL, Leinenbach C, Vassiliou MF, and Carstensen JV. Additive Manufacturing for Fast Prototyping of Topology Optimized Designs of Reinforced Concrete. *Additive Manufacturing*.

[P1] **Elmorsy M**, Jones L, Cazacu M, Anastasopoulos I, and Vassiliou MF. Centrifuge Shake Table Tests on Rolling Isolated Masonry Model Structures. *Engineering Structures*.

CONFERENCE PAPERS

[C12] **Elmorsy M**, Jones L, Wild M, Katsamakas A, Anastasopoulos I, and Vassiliou M. Centrifuge Modelling of URM Seismic Response Using Sand-Based 3D Printing. In *11th International Conference on Physical Modelling in Geotechnics (ICPMG 2026)*. Zurich, Switzerland. 2026. doi: 10.53243/ICPMG2026-434

[C11] **Elmorsy M**, Leinenbach C, and Vassiliou MF. Physical Modeling of Reinforced Concrete Structures with 3D-Printed Steel Cages at Centrifuge Scale. In *11th International Conference on Physical Modelling in Geotechnics (ICPMG 2026)*. Zurich, Switzerland. 2026. doi: 10.53243/ICPMG2026-435

[C10] **Elmorsy M**, Wrobel R, Leinenbach C, and Vassiliou MF. Applications of 3D Printing for Manufacturing Small-Scale Physical Models of Reinforced Concrete Structures for Centrifuge Use. In *6th Panhellenic Conference on Earthquake Engineering and Seismological Techniques*. Athens, Greece. 2025. (*In Greek*). doi: 10.3929/ethz-c-000788023

[C9] **Elmorsy M,** Leinenbach C, and Vassiliou MF. Assessment of 3D Printed RC Beam-Column Joints Models for Centrifuge Modelling. In *1st Hellenic-Italian Conference on Computational Mechanics, Biomechanics and Mechanics of Materials (HICOMP 2025)*. Rhodes Island, Greece. 2025. doi: 10.3929/ethz-c-000743504

[C8] **Elmorsy M**, Katsamakas A, Jones L, Brunschweiler E, Anastasopoulos I, and Vassiliou MF. Seismic Behavior of URM Structures: A Centrifuge Model Study. In *International Conference on Structural Analysis of Historical Constructions (SAHC 2025)*. Lausanne, Switzerland. 2025. doi: 10.1007/978-3-032-16767-5_54

[C7] **Elmorsy M**, Katsamakas A, Jones L, Brunschweiler E, Anastasopoulos I, and Vassiliou MF. Preliminary Centrifuge Tests on Unreinforced Masonry Buildings Built Using a Sand-Based 3D Printer. In *4th Asia-Pacific Conference on Physical Modelling in Geotechnics (ACPMG 2024)*. Abu Dhabi, UAE. 2024. doi: 10.53243/ACPMG2024-69

[C6] **Elmorsy M**, Leinenbach C, and Vassiliou MF. Enhancing Centrifuge-Based Physical Modeling of Reinforced Concrete with 3D Printed Reinforcement Cages. In *4th Asia-Pacific Conference on Physical Modelling in Geotechnics (ACPMG 2024)*. Abu Dhabi, UAE. 2024. doi: 10.53243/ACPMG2024-70

[C5] **Elmorsy M**, Wrobel R, Leinenbach C, and Vassiliou, MF. Centrifuge-Based Physical Modeling of Reinforced Concrete Using 3D Printed Reinforcement Cages. In *5th European Conference on Physical Modelling in Geotechnics (ECPMG 2024)*. Delft, Netherlands. 2024. doi: 10.53243/ECPMG2024-141

[C4] **Elmorsy M**, Wrobel R, Leinenbach C, and Vassiliou, MF. Small scale physical modeling of reinforced concrete using 3D printed steel reinforcement. In *18th World Conference on Earthquake Engineering (WCEE 2024)*. Milan, Italy. 2024. doi: 10.3929/ethz-b-000679333

[C3] **Elmorsy M** and Vassiliou MF. Do ground motion processing and filtering affect the behavior of rocking structures? In *18th World Conference on Earthquake Engineering (WCEE 2024)*. Milan, Italy. 2024. doi: 10.3929/ethz-b-000679334

[C2] **Elmorsy M** and Vassiliou MF. Ground motion processing in rocking structures. In *9th ECCOMAS Thematic Conference on Computational Methods in Structural Dynamics and Earthquake Engineering (COMPDYN 2023)*. Athens, Greece. 2023. doi: 10.7712/120123.10718.20776

[C1] **Elmorsy M**, Wrobel R, Leinenbach C, and Vassiliou MF. Material Testing of Micro-Concrete and 3D-Printed Reinforcement for Use in Small-Scale Seismic Testing of RC Structures. In *9th ECCOMAS Thematic Conference on Computational Methods in Structural Dynamics and Earthquake Engineering (COMPDYN 2023)*. Athens, Greece. 2023. doi: 10.7712/120123.10696.20779

CONFERENCE PRESENTATIONS

[T3] **Elmorsy M**, Vassiliou M, and Vamvatsikos D. Propagation of RC Beam-Column Joint Modeling Uncertainty to the Seismic Performance of RC Buildings Using Incremental Record-Wise LHS. In *Engineering Mechanics Institute Conference and Probabilistic Mechanics & Reliability Conference (EMI/PMC 2024)*. Chicago, IL, USA. 2024.

[T2] **Elmorsy M** and Vassiliou MF. Validation of System-Level Assumptions in Seismic Analysis of RC Structures: An Experimental Framework Based on 3D Printing of the Reinforcement. In *Engineering Mechanics Institute Conference and Probabilistic Mechanics & Reliability Conference (EMI/PMC 2024)*. Chicago, IL, USA. 2024.

[T1] **Elmorsy M** and Vassiliou MF. An Experimental Framework for Statistical Validation of System Level Assumptions in Seismic Analysis of RC Structures. In *21st Swiss Geoscience Meeting*. Swiss Academy of Sciences. Mendrisio, Switzerland. 2023.

DATASETS

[D1] **Elmorsy M,** Jones L, Wild M, Tatsis K, Anastasopoulos I, and Vassiliou MF. Dataset from shake table centrifuge tests of 3D printed unreinforced masonry structures. *ETH Research Collection*. Dataset. 2025. doi: 10.3929/ethz-b-000745078

INVITED TALKS

(2026) **Making Structural Engineering More Predictive, More Sustainable, and More Accessible**. (2026). EERI Journal Club. EERI Student Chapters Polytechnique Montréal, Purdue University, and McGill University. Online (Scheduled).

(2026) **Additive Manufacturing for Benchmarking and Uncertainty Reduction in Structural Engineering**. iMMEC internal Mechanics Seminar: Modeling, Experimentation, Computation. Massachusetts Institute of Technology (MIT), Cambridge, MA.

(2025) **Validation of System-Level Assumptions in Seismic Analysis of RC and Masonry Structures: An Experimental Framework**. ETH Zurich Geotechnical Centrifuge Center Inauguration Event. ETH Zurich, Zurich, Switzerland.

MENTORING EXPERIENCE

ETH Zurich Zurich, Switzerland

- **Vasile Madalin Cazacu. Master Thesis:** Experimental design and seismic isolation techniques for small-scale masonry structures using innovative bearing systems (in collaboration with DTU, Denmark). Thesis results contribute to [P1], on which he is a co-author. Fall 2024
- **Manuel Wild. Master Thesis:** Centrifuge modeling and additive manufacturing techniques for masonry structures using binder jetting 3D printing. His thesis contributed to [J12] and [C12], on which he is a co-author. Fall 2024
- **Stefan Murer, Julian Schlachter, and Manuel Wild. Master Project:** Development of 3D-printed reinforcement cages and experimental techniques for small-scale RC modeling in centrifuge testing Fall 2023
- **Morena Giulieri. Master Project:** Physical modeling approaches for reinforced concrete and soil–structure interaction in centrifuge experiments Fall 2022

Mansoura University (2017-2019)

Capstone Graduation Projects: Mentored undergraduate student teams on structural design projects, providing guidance on structural analysis, design, modeling, and technical presentation

TEACHING EXPERIENCE

ETH Zurich Zurich, Switzerland

Graduate Teaching Assistant Feb. 2023 – Jan. 2024

- Structural Dynamics and Vibration Problems (Graduate), Seismic and Vibration Isolation (Graduate); supported instruction for classes through office hours and homework model answers

Mansoura University Mansoura, Egypt

Teaching Assistant Jan. 2017 – Dec. 2019 & Jan. 2021 – Dec. 2021

- Design of Concrete Structures I, II, and III (Undergraduate), Computer Applications in Structural Engineering (Undergraduate); conducted weekly tutorial sessions, graded homework and exams, and offered office hours

RELEVANT PROFESSIONAL EXPERIENCE

Hamed Askar Consulting Office Mansoura, Egypt

Structural Design Engineer (part-time) Aug. 2016 – Jan. 2018

- Performed structural analysis and design of reinforced concrete frames for residential buildings
- Designed storage tanks for water treatment facilities
- Conducted wind and seismic load analyses for structural designs

ECG Engineering Consultants Group S.A. Cairo, Egypt

Intern July 2015 – Oct. 2015

- Training in all phases of structural design alongside experts at a leading consulting firm in the Middle East
- Full design and detailing of a 30-story high-rise building (1st place in the end of internship competition)

PROFESSIONAL SERVICE

Reviewer, *ASCE Journal of Structural Engineering; Bulletin of Earthquake Engineering; Engineering Structures; ASCE Journal of Engineering Mechanics; Journal of Architectural Engineering*

Member, *Organizing Committee, PEER Blind Prediction Competition, Pacific Earthquake Engineering Research Center, UC Berkeley, 2025.* Link

Volunteer, *Organizing Committee, International Association for Shell and Spatial Structures (IASS) Symposium, Zurich, Switzerland 2024*

PROFESSIONAL ASSOCIATIONS

Earthquake Engineering Research Institute (EERI); American Society of Civil Engineers (ASCE)

REFERENCES

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Department of Civil and Environmental Engineering
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Postdoctoral PI

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PhD Second Advisor and Collaborator

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Collaborator